AI, Art, and Ethics

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With the rapid expansion of the use of generative AI, it is important to understand the implications of this emerging technology in different domains and sectors. In this module, Srinivasan explores the ethical challenges that emerge from the use of generative AI in the field of visual art, often referred to as generative art. Through a series of case studies, Srinivasan highlights existing biases in the field and shares specific tools that can drive the responsible development of art datasets and the use of ethical frameworks to assess the impact of AI systems.

Lecture Transcript

0:00 Hello everyone. My name is Ramya Srinivasan. I'm a research scientist and manager at Fujitsu Research of America. In this talk, I will be touching upon some topics at the intersection of AI, art and ethics. Specifically, I will be speaking on biases in generative art, responsible development of art datasets, and more, and more broadly on frameworks to assess the ethical impact of AI systems. So let's get started.

0:33 The very first question that arises is why even study AI, art and ethics? The simple answer is that there is an increased use of AI for art and creativity, which necessitates us to study these topics. Machine learning technologies are being rapidly integrated into the creative arts domains, as evidenced by the number of new art data sets, generative art galleries, and applications concerning the art. Although motivated by new artistic practices, ML assisted or ML based digital art and associated technologies can bring with them ethical concerns ranging from representational harm to cultural appropriation.

1:18 Let's consider the case of generative art as an instance. Generative art, as one might know, commonly refers to art that is created in part or in whole by an automated system such as the AI system here. Generative art can perform a variety of tasks, such as creating a portrait from a photograph, as you can see in this figure. On the left, you can see the photo of actress Tessa Thompson. And the right image is a portrait of the actress as generated by an app called Aiportraits. As can be noticed, the skin color of the actress is lighter in this version. This was something that was observed across all images fed to this app, and not just specific to this actress. Upon investigation, it was found that this was due to the fact that the app itself was trained on about 15,000 white Renaissance portraits. So clearly biases in training data and algorithms could result in discrimination based on sensitive attributes such as race, gender, age, and so on.

2:41 More broadly, biases in generative art can have long standing adverse socio-cultural impacts. Historical events and people may be depicted in a manner contrary to the original times. And owing to automation bias, whereby people tend to believe the results of an algorithm more than what actual historical evidence is pointing to, there could be a bias in understanding history, and this could hinder in the preservation of cultural heritage.

3:13 In our paper "Biases in Generative Art – A Causal Look from the Lens of Art History", a paper that was published in ACM FAccT 2021. We systematically investigated biases in the generative art pipeline, right from those that can arise due to improper problem formulation to those related to algorithm design and evaluation. Specifically, viewing from the lens of art history, we discussed the socio-cultural impacts of such biases and highlighted how correct methods fall short in modeling the process of art creation. In order to do so we leveraged directed acyclic graphs or DAGs and illustrated the different types of biases through case studies. So I will now briefly discuss some case studies and results from this paper. And before I do so, I need to go over certain background materials or terms that I will be using in the subsequent slides. Some of these are related to the art community and some related to the causal machine learning communities. So I will go over them one by one. Although some of these terms may be familiar, I'm just going to define them for completeness.

4:24 The first one: art movement. These refer to tendencies or styles in art, which have a common specific philosophy, which is influenced by various factors such as culture, geography, political-dynastical markers, and so on. Typically, this is followed by a group of artists during a specific period of time. As an example, on the very right here is an artwork belonging to Impressionism, which is a modern art movement belonging to the 19th century, and you can see artworks belonging to different or older art movements in the center and left respectively. Genre may be a term that everyone knows. This basically depicts the theme or the object in the artwork, like landscapes, portraits, paintings, these are different genres. Art material refers to the substance used to create the artwork like oil, charcoal, clay, marble, painting and so on. The term artist style is a very abstract and rather contested term. So in addition to these factors, such as art movement, genre and art material, there are several other factors that can characterize an artist's style. These could be cultural factors, such as the artist's background, their art schools or lineage and also many subjective elements such as their cognitive skills, beliefs, prejudices, upbringing and so on. So clearly, modeling artist style computationally is a very complex and contested problem, and several biases can result from this process. We will look into this in the subsequent slides as well.

6:19 Now let's define some terms from the causal machine learning community that we will be using here. Directed acyclic graphs which I referred to earlier, these are basically visual graphs, which are used for encoding causal assumptions between variables of interest. What are our variables of interest? Artists, of course, the art material, artwork, genre art movement, these are all our variables of interest. And the directed acyclic graph basically captures the causal dependencies between these variables of interest. Here you can see one such DAG, here X represents the artist, Z the art work. So, the arrow from X to Z is in some sense capturing the style of the artist, the influence of the artist on the artwork right? And in the process, several factors can influence. For example, the genre (G) can influence how the artist creates the artwork and how the artwork is rendered. So, there is an arrow from G to X as well as G to Z. Similarly, the art movement (A) can influence both the artist and the art material. So there is an arrow from A to X and A to Z. Art material (M) also can influence both the artist and the artwork. So there is an arrow from M to X and M to Z. Further the art movement itself can influence the art material. For example, marble was the most common art material used during the Renaissance period. So, the art movement can influence the art material. There is an arrow from A to M. And this is just one set of assumptions based on some expert opinion.

8:03 And based on different expert opinions, there can be multiple DAGs and it is possible to analyze for biases across all such scenarios. So basically, DAGs serve as a tool to test for various biases under such assumptions depicted in the DAG, and it facilitates domain experts such as art historians to encode their assumptions, and hence, serve as accessible visualization and analysis tools. Here we outline how we actually selected the case studies for our analysis. Basically, we surveyed academic papers, online platforms and tools that claim to generate established art movement styles and artist's styles. And this table here lists the various art movements, artists, genres and materials that were analyzed in the paper. I will now discuss a few case studies. I won't be able to discuss all the case studies illustrated in the paper due to lack of time, so I will just discuss a few of them.

9:11 We earlier saw the case of Aiportraits, which was largely trained on white Renaissance men due to which the rendered portraits were all lighter in the color–an incidence of racial bias. So, this is an instance of selection bias more specifically. Here, in that example, the type of artworks used in the dataset created that bias and that can be succinctly captured through the DAG. Now in this DAG here, X again represents the artist, Z represents the artwork. Now there is a special variable denoted by the concentric circle and S here, which is called as the selection variable. This basically is used to indicate if there is a factor that influences the selection of that particular variable. In the case of Aiportraits, we saw that the type of artworks that were selected were mostly those of white Renaissance men, so there is an arrow from Z to S here to denote that particular selection. Now, I won't be able to discuss the details of how these kinds of DAGs and their encodings translate in terms of identifying the bias, but basically, there are a set of conditions known as selection backdoor criteria that need to be satisfied in order to avoid selection bias. And if there is a path from the selection variable, sorry, between the selection variable and the output of interest, then the selection backdoor criteria says that there is selection bias.

11:04 In this case, there is a connection between Z and S, which is the artwork, our output variable of interest, and the selection variable. And therefore, one can conclude that there is selection bias. So we can see how DAGs can serve as accessible means to analyze for bias without having to look through the dataset, or the results. By just understanding how this data set was curated, it is possible to say that there is selection bias in the process. So DAGs are extremely accessible and can facilitate non-AI people or domain experts to verify their hypotheses.

11:52 Let's consider another illustration of bias in the generative art pipeline. Here we will discuss style transfer bias. Before we understand this, we need to know what is style transfer? Style transfer refers to the problem of converting the style of one artwork into the style of another artwork. Here, the style could refer to the artist's style, it could be the style of the art movement, or changing the art material itself, and so on. So an example will clarify this better. So let's look at this figure here. The center image in this figure is an artwork belonging to the Cubism art movement, which is a modern art movement. It is an artwork titled "Propellers" by artist Fernand Leger. The Industrial Revolution was just setting in during that time when Cubism was in place. And so it was common for artists of those times to depict the technological advancements of those times in their artwork. So here you can see one such technological advancement that is the propeller. On the very right you can see a Futurism artwork. Futurism is also another modern art movement. Here, artists emphasized on kinetic patterns in their artwork, and most artists voiced their opinions about the political movements and wars that were prevalent during those times through their artworks. In fact, this particular art piece you can see here, "Armoured train in Action" is what it is called, is a depiction of a train carrying soldiers and gusting through the countryside. So you can see the wind coming out of this train as well. So that's an instance of the kinetic patterns that I was referring to earlier.

13:59 On the very left is a transfer or translation of the center image according to the style of the right image. So basically, the leftmost image is trying to transfer the center image which is in Cubism art movement, into Futurism, which is depicted in the very right and this is by a platform called DeepArt.io. As you can see, the transferred image shown on the very left is somewhat similar in shade and color to the rightmost image. But the main factor of Futurism art movement, namely the kinetic patterns, is missing here. That was a very key aspect of Futurism, the kinetic patterns, that is missing in this transformed version. You can, one could have easily imagined showing these kinetic patterns associated with propellers, but that is not evident in this transferred version. So, this is a clear instance of another bias wherein the intricate details of the art movements are missed. And therefore, aspects of history, and cultural and social contexts are lost in the process. This is exactly what I was referring to earlier, when I was talking about the concern surrounding generative art.

15:38 Again this can be captured through the DAGs as one can see here on the right. X represents here the artwork belonging to the Cubism movement, and Z, the artwork belonging to the Futurism movement. And one has to recognize that there could be several unobserved factors, such as the artists who have created these artworks, the art movements themselves, the genres, and other aspects like the art material, which could also induce differences across these renditions. So all of that is captured through this variable Y, which is shown in a dotted circle. It is dotted to indicate that this is unobserved. We don't know the artist or the other factors that would have influenced these artworks and artists. So apart from that, there are these square blocks denoted by R. These are also variables. They basically denote the differences between these two art movements, genres, and all that. These are special variables similar to the selection variables. They also have special connotations. I wouldn't be able to discuss those in detail. But basically, there are a set of criteria which can say when the causal effect of one variable can be transferred to another. And under these circumstances depicted in the DAG, it can be inferred that the causal effect on X cannot be transferred to Z. So looking at the DAG, it is possible to infer whether the causal effects can be transferred across variables or not. Again, this can be inferred by looking at the DAG. And we also analyzed it looking at the images in more detail. So this is another instance where DAGs can facilitate analysis of such biases.

18:00 Here are some more illustrations of style transfer bias. Images one (i) and three (iii) depicted here are real portraits of young men, by artists Raphael and Cosimo respectively. And two (ii) and four (iv) are gender translations of one and three respectively, by a platform called abacus.ai. So basically, gender translations, as done by abacus.ai is that men are converted into women, women into men, and so on. So the platform mistakes these portraits one and three to be those of women. Actually, these are portraits of young men but because these portraits have images of people having hair up to their shoulders, the platform thinks it is a female, and therefore the generated versions show mustache and beard and so on. But young men, in those times, in the Renaissance times, had long hair often extending beyond their ears up to their shoulders. So, these kinds of subtle cultural contexts are missed in these platforms, which again result in wrong style transfers. Again, another instance of cultural bias here.

19:42 Another instance of style transfer bias, more specifically, racial bias can be seen in this illustration. Here, one (i) and three (iii) are real artworks. Specifically one is a folk art by artist Clementine Hunter. Three is a Renaissance sculpture. Two (ii) and four (iv) are generated versions of one and three respectively according to the Expressionism art movement. Expressionism is another art movement. And without going into the details of the Expressionism art movement, you can just look at these figures to see how the face color is actually changed. The difference in the face color between one and three is you can observe. The black portrayal for folk art is rendered into a red face in two, whereas the white sculpture remains more or less white in the generated version four. So this is another instance of racial bias as you can see.

20:54 So we saw there were several gaps across these illustrations, different types of biases. We saw selection bias, racial bias, cultural bias, and so on. So how do we address such gaps? Clearly, there is a need to analyze socio-cultural impacts of generative art tools before their deployment. We need to question whether an algorithm can even solve a given problem. For example, can the style of an artist be modeled? Next, we need to define guidelines related to accountability of generative art as well. There is a need to involve domain experts in the generative art pipeline to better inform the process of art creation. And we need to also outline the details of generative models to facilitate better communication between researchers and developers of such models.

21:50 Towards addressing some of these gaps that we discussed, we designed a checklists called as Artsheets to offer dataset creators with a framework to facilitate critical reflection of various aspects of their work, and to offer guardrails for types of art based datasets now in popular use by the ML community. In order to guide users to make an artsheet, our checklist is divided into thematic sections. This workflow leaves the users to consider the ethical issues from project ideation through executions to plans for maintenance across time. The thematic sections include motivation, in other words, the dataset's creator's purpose and contextual conditions, data provenance, data composition, data collection, data pre-processing, cleaning and labeling, use and distribution, data generation and data maintenance. This slide here shows sample questions pertaining to our data collection, pre-processing, use and distribution.

23:09 Our framework can also be leveraged as a tool for retrospective and historical dataset analysis and documentation. This slide here summarizes the findings from two case studies of art sheets applied retrospectively to ArtEmis, a visual art dataset, and JF Fake Chorales, a generated music dataset. The analysis shows that non-western and indigenous art forms are underrepresented. There is inadequate provenance information and insufficient documentation pertaining to annotator backgrounds. For details, you can refer to our paper, which is listed here in this slide

24:00 We, at Fujitsu Research, are committed to delivering trustworthy AI and towards realizing this vision we have released an AI Ethics Impact Assessment Tool. This framework incorporates information obtained by interaction between stakeholders and the AI system to provide a user friendly risk analysis diagram. AI Ethics Impact Assessment tool, or AIEIA, can identify the ethical implications of AI systems comprehensively, compliant with ethical guidelines. A whitepaper describing AIEIA was published in February 2022 and is available for the public. So with that, I would like to close this presentation. For more details, one can refer to these references listed here. Thank you for listening.

